

MAT 1341C – DGD 1 – Test 4, 2015

30-March, 2015.

Instructor: Barry Jessup.

Family Name:_____

First Name:_____

Student number:_____

Multiple choice answers →

For the marker's use only →

1	
2	
3	
subtotal	
4	
5	
6	
[Bonus] 7	
Total	

PLEASE READ THESE INSTRUCTIONS CAREFULLY.

1. Read each question carefully, and **answer all questions in the space provided after each question.** For questions 4 to 7, you may use the backs of pages if necessary, but be sure to indicate to the marker that you have done this.
2. Questions 1 to 3 are multiple choice. These questions are worth 1 point each and no part marks will be given. Please record your answers in the space provided above.
3. Questions 4 - 6 are worth 6 points each, and part marks can be earned in each. **The correct answers here require justification written legibly and logically: you must convince the marker that you know why your solution is correct.** Question 7 is a bonus question and is worth only 4 points. It is much more difficult to earn points in the bonus question.
4. Where it is possible to check your work, do so.
5. Good luck! Bonne chance!

1. If G is an $n \times 2$ matrix and $H = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ then the second column of the matrix GH is

A. not defined unless $n = 2$.
B. the same as the second column of G.
C. the same as the second column of H.
D. the same as the first column of G.
E. the same as the first column of H.
F. the sum of the first and the second column of G.

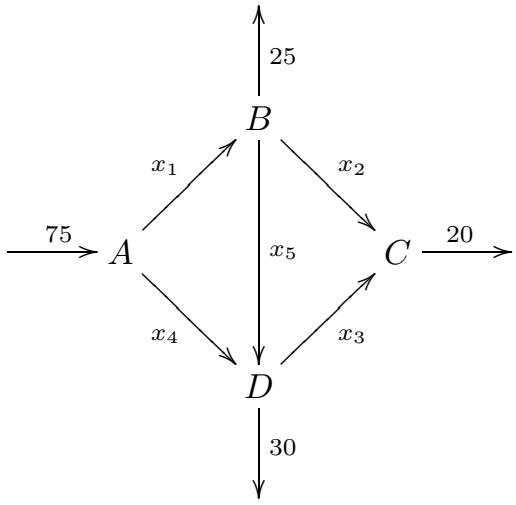
2. If $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$, then the third row of B^{-1} is:

A. $[0 \;\; 1 \;\; 0]$
B. $[1 \;\; 0 \;\; 2]$
C. $[-1 \;\; 1 \;\; 1]$
D. $[0 \;\; 0 \;\; 1]$
E. $[1 \;\; 0 \;\; -1]$
F. $[1 \;\; 0 \;\; 1]$

3. What is the dimension of the subspace of $\mathbf{R}^5$ spanned by $(1, -1, 4, -5, 0), (2, 1, 5, -1, 0)$, and $(0, 1, -1, 3, 0)$?

A. 0
B. 1
C. 2
D. 3
E. 4
F. 5

4. Consider the network of streets with intersections A, B, C and D below. The arrows indicate the direction of traffic flow along the one way streets, and the numbers refer to the number of cars observed to enter A or leave B, C and D during one minute. Each x_i denotes the unknown number of cars which passed along the indicated streets during the same period.



a) Write down the linear system which describes the the traffic flow, **together with all the constraints** on the variables x_i, $i = 1, \ldots, 5$. (Do not perform any operations on your equations: this is done for you in (b).)

4b) The reduced row-echelon form of the augmented matrix from part (a) is

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 0 & 75 \\ 0 & 1 & 0 & 1 & 1 & 50 \\ 0 & 0 & 1 & -1 & -1 & -30 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general solution. (Ignore the constraints at this point.)

c) If BD were closed due to roadwork, find the minimum flow on AB, using your results from (b).

5. Let $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 5 & 10 & -4 & 19 \\ 2 & 4 & -2 & 8 \end{bmatrix}$.

a) Find a basis for the row space $\text{row}\,(A)$ of A.

b) Find a basis for the column space $\text{col}\,(A)$ of A.

c) Find the dimension of $\{Ax \mid x \in \mathbf{R}^4\}$.

d) Find a basis for $\ker(A) = \{x \in \mathbf{R}^4 \mid Ax = 0\}$.

6. (a) State whether each of the following statements is (always) true, or is (possibly) false, in the box after the statement.

- If you say the statement may be false, you must give an explicit example - with numbers!
- If you say the statement is always true, you must give a clear explanation.

i) If A is a 3 by 4 matrix, then the rows of A must be linearly dependent.

ANSWER

ii) If A is a 2 by 2 matrix and $A^2 = 0$, then $A = 0$.

ANSWER

6 (b). Let A be a real $n \times n$ matrix. Give 3 statements (in total) equivalent to

"A is invertible",

one each in terms of:

(I) the columns of A

(II) the reduced row-echelon form of A

(III) the homogeneous system $Ax = 0$.

7. [Bonus] Prove carefully that if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ satisfies $ad = bc$, then A is not invertible. (Note: you cannot use the formula for the inverse of a 2 by 2 matrix, since the formula relies on $ad \neq bc$.)

(You may use this page for rough work or solutions that did not fit on previous pages.)